

HANOVER COUNTY MUNICIPAL, VA, USA

WMO: 723084

Lat: 37.708N

Lon: 77.434W

Elev: 205

StdP: 14.59

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 93775

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 15.9	(c) 19.3	(d) -1.7	(e) 5.0	(f) 22.9	(g) 2.9	(h) 6.4	(i) 25.9	(j) 20.8	(k) 50.2	(l) 18.8	(m) 47.4	(n) 3.0	(o) 290	(p) 0.405

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	94.9	75.3	92.4	74.7	90.1	74.2	78.3	89.2	77.2	87.7	76.1	85.8	7.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.2	133.4	81.7	74.3	128.9	80.8	73.2	124.1	80.0	41.7	89.9	40.6	87.6	39.6	85.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 17.8	(b) 15.5	(c) 12.7	(e) 7.2	(f) 98.1	(g) 5.7	(h) 3.3	(i) 3.1	(j) 100.4	(k) -0.2	(l) 102.3	(m) -3.4	(n) 104.2	(o) -7.5	(p) 106.6		
(4) 17.8	(5) 15.5	(6) 12.7	(7) 7.2	(8) 98.1	(9) 5.7	(10) 3.3	(11) 3.1	(12) 100.4	(13) -0.2	(14) 102.3	(15) -3.4	(16) 104.2	(17) -7.5	(18) 106.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.5	37.4	40.4	48.0	58.2	66.8	74.6	78.4	77.1	70.4	59.6	49.1	41.3		
	DBStd	16.56	10.77	9.77	9.98	8.99	7.80	5.92	4.94	4.91	6.38	8.41	8.92	9.27		
	HDD50	1338	414	299	163	23	0	0	0	0	0	14	123	302		
	HDD65	3853	855	689	533	243	77	4	0	0	24	213	480	735		
	CDD50	4457	24	31	100	270	521	739	881	839	612	310	97	33		
	CDD65	1496	0	1	5	40	132	293	416	375	186	44	3	1		
	CDH74	14340	1	7	75	472	1252	2882	4266	3528	1472	356	24	5		
	CDH80	5804	0	0	10	135	404	1190	1971	1517	491	85	1	0		
(14) Wind	WSAvg	4.7	5.4	5.6	6.0	6.0	4.9	4.5	4.0	3.5	3.5	4.0	4.4	4.6		
(15) Precipitation	PrecAvg	43.6	3.3	2.8	4.0	3.3	3.9	4.0	4.3	4.2	4.3	3.1	2.9	3.6		
	PrecMax	64.1	6.9	6.3	8.5	8.1	9.7	7.7	8.3	14.2	15.0	6.6	10.0	8.5		
	PrecMin	33.2	1.2	0.9	0.2	1.2	1.0	1.0	1.2	1.4	0.0	0.0	0.0	0.9		
	PrecStd	7.1	1.3	1.3	1.8	1.5	2.0	1.8	1.7	2.5	3.2	1.8	2.1	1.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.9	74.1	81.2	88.3	91.0	96.7	99.2	97.4	93.0	87.3	77.4	73.0		
		MCWB	60.5	60.6	61.8	66.8	72.6	74.7	76.3	75.9	73.2	71.3	64.8	62.0		
	2%	DB	65.8	68.1	75.2	83.9	87.8	92.7	95.3	93.5	89.2	81.8	72.6	67.1		
		MCWB	57.7	57.2	59.8	64.5	71.1	74.6	76.0	75.0	72.9	68.2	61.1	60.8		
	5%	DB	61.4	63.1	70.9	80.0	84.4	90.1	92.6	91.0	85.5	78.1	69.3	62.8		
		MCWB	54.1	52.5	57.2	63.1	70.0	74.1	75.2	74.7	71.1	66.9	60.4	56.0		
	10%	DB	55.1	57.3	65.8	74.9	81.4	87.3	90.0	88.3	82.0	73.7	64.4	57.1		
		MCWB	47.7	48.7	54.6	61.3	68.2	72.7	74.9	73.8	70.1	64.9	57.2	50.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.8	64.0	64.9	70.0	75.5	78.4	80.3	79.5	77.7	75.1	68.0	66.5		
		MCDB	67.8	70.8	74.6	80.8	85.3	88.1	90.2	91.4	85.2	80.5	72.6	69.7		
	2%	WB	59.6	59.3	62.3	67.5	73.3	76.8	78.4	78.0	75.7	72.2	65.1	61.6		
		MCDB	64.9	66.0	71.9	79.2	83.4	87.5	89.9	89.2	83.4	78.5	70.0	65.6		
	5%	WB	55.0	54.6	59.6	65.3	71.7	75.6	77.4	76.9	74.4	69.4	62.0	57.2		
		MCDB	60.0	60.5	68.0	76.2	81.4	86.1	88.5	86.8	80.9	75.7	66.7	61.4		
	10%	WB	48.4	49.6	56.4	63.2	70.0	74.5	76.3	75.6	72.8	66.3	58.7	51.7		
		MCDB	53.8	57.2	64.2	72.5	78.9	84.1	86.4	84.0	78.7	72.0	64.2	55.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.8	20.8	22.3	23.4	21.3	20.6	20.0	19.7	19.9	21.1	21.4	19.4		
		MCDBR	25.2	27.4	28.0	28.1	23.9	23.4	23.2	22.8	23.4	24.3	25.0	24.4		
		MCWBR	18.7	18.3	15.7	12.6	10.9	9.1	8.1	7.7	9.4	12.4	15.2	18.1		
	5% WB	MCDBR	23.2	24.5	24.5	23.8	20.6	20.5	20.1	19.1	18.0	19.5	19.7	21.6		
		MCWBR	19.9	19.2	16.6	12.3	10.0	8.9	8.1	7.7	8.7	11.6	14.3	18.5		
(40) Clear-Sky Solar Irradiance	taub		0.300	0.314	0.347	0.389	0.427	0.485	0.503	0.505	0.408	0.352	0.319	0.304		
	taud		2.529	2.477	2.426	2.322	2.242	2.124	2.097	2.073	2.330	2.485	2.535	2.553		
	Ebn at Noon		283	292	291	282	272	256	250	246	267	274	273	274		
	Edh at Noon		24	29	34	40	44	49	50	50	37	29	24	23		
(44) All-Sky Solar Radiation	RadAvg		723	968	1282	1611	1828	1991	1905	1718	1402	1092	816	636		
	RadStd		61	108	101	124	173	140	97	83	150	128	65	57		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (74 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+247	+151

Nomenclature: See separate page